
Harder Instances, Higher Accuracy: Difficulty, Oracles, and Item Dependence in LLM Robustness Benchmarks

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Abstract

Benchmarks that perturb the surface form of reasoning problems report accuracy drops and attribute them to memorisation. We audit what those drops measure. Applying six surface variants to 219 problems across arithmetic, planning, and algorithmic optimisation, on six models, we find that three confounds each independently produce drop-shaped signals. First, the answer-checker is itself a program: because obfuscation perturbs exactly the tokens a checker parses, verifier design is correlated with the treatment rather than random, and it biases toward the memorisation hypothesis. Holding responses fixed and removing one verifier component moves a planning cell from 0.252 to 0.043 with zero movement on the unperturbed condition. Second, canonical and regenerated instances are not difficulty-matched. On Blocksworld, regenerated instances are harder on every structural measure we can compute, including optimal plan length from an admissible planner, yet four of six models score *higher* on them; Claude Sonnet 4 moves from 0.170 to 0.638 unmatched. Stratifying on block-naming convention shrinks that gap to within-stratum deltas of +0.090 and +0.064 (matched naming $n=13$: +0.077), so the headline inversion is confounded by lexical form as well as difficulty. n -gram proximity to public corpora does not account for the residual gap. Third, benchmark items are not independent: our algorithmic bank of 110 problems collapses to 51 clone families, so nominal intervals are $1.6\text{--}2.7\times$ too narrow. Beyond the confounds, two structural results survive. Surface fragility has no item-level locus — no problem fails across all models in arithmetic or algorithms — and plan-execution consistency covaries with rename survival when pooled across families ($r=0.243$, $p=0.024$, $n=272$, cluster bootstrap) while remaining distinct from raw accuracy. We release per-instance verifier logs and a gold-in-gold-out harness that gates verifier validity before any model is queried.

1 Introduction

A now-standard evaluation protocol takes a reasoning benchmark, perturbs the surface form of each item while preserving its solution, and reports the accuracy drop as evidence that models rely on memorised surface patterns rather than generalisable procedure [Mirzadeh et al., 2025, Valmeekam et al., 2023, Wang and Zhao, 2024, Huang et al., 2025]. The inference is attractive because the manipulation is clean: if the problem is logically identical and only the words changed, a drop looks like direct evidence about representation.

This paper asks what else could produce that drop. We are not arguing the phenomenon is absent. We are arguing that the standard protocol, as implemented across several widely-used benchmarks,

admits at least three alternative explanations that are rarely controlled and that each bias in the same direction as the hypothesis under test.

The oracle is a program. Automated evaluation depends on an answer-checker, and in software testing the reliability of that checker is known as the *test oracle problem* [Barr et al., 2015]. Obfuscation benchmarks have an unusually severe version of it. The perturbation renames exactly the entities, actions, and identifiers that a checker must parse to score an answer. A checker that silently fails to invert the rename returns *incorrect* on the perturbed condition and *correct* on the canonical one, for every model. This is not measurement noise: the failure is correlated with the treatment, so it produces bias, and the bias points toward the memorisation hypothesis. Section 4 ablates three verifier components under fixed responses and shows that the measured drop is verifier-dependent.

Regenerated instances are not matched on difficulty or lexical form. Comparing a benchmark’s released items against freshly generated ones from the same domain is a natural exposure control. It is only valid if the two sets are matched on difficulty *and* surface conventions. On Blocksworld we compute optimal plan length with an admissible planner and find the regenerated set is harder on structural axes — yet the unmatched accuracy inversion is largely attenuated once block-naming convention is stratified (Section 5). Uncontrolled difficulty can manufacture a drop, uncontrolled lexical form can manufacture an inversion, and either can hide the other.

Items are not independent. Procedurally generated banks produce clone families. Our 110-problem algorithmic bank contains 14 families covering 73 problems, giving an effective n of 51. Confidence intervals computed under an independence assumption are $1.6\text{--}2.7\times$ too narrow (Section 6). We are not aware of a rename-robustness paper that reports effective n .

Contributions.

- A difficulty- and naming-stratified comparison of benchmark-released versus procedurally regenerated planning instances, using admissible-planner optimal plan length as a structural difficulty measure and documenting that the unmatched accuracy inversion is confounded by block-naming convention (matched $n=13$).
- A verifier-ablation characterisation of treatment-correlated scoring sensitivity in obfuscation designs, with a gold-in-gold-out gate and three documented component toggles with their magnitude and sign.
- Evidence that surface fragility has no item-level locus and that plan-execution consistency shows convergent association with rename survival when pooled, while remaining discriminant from raw accuracy.
- Per-instance verifier logs, exclusion codes with reasons, and clone-family annotations for a 219-problem bank across three domains.

2 Related Work

Surface-form robustness. Mirzadeh et al. [2025] introduce GSM-Symbolic, which regenerates GSM8K items with varied names and values and reports group-level accuracy decline; their *Vary Name* condition is the direct predecessor of our entity-rename variant on arithmetic. Valmeekam et al. [2023] introduce PlanBench, whose Obfuscated Mystery Blocksworld replaces domain predicates with arbitrary tokens; our planning rename variant uses that design and, for the mystery subset, that vocabulary. Wu et al. [2024] extend counterfactual evaluation across eleven tasks, Wang and Zhao [2024] apply nine perturbation types across fifteen datasets, Li et al. [2024] generate GSM8K variants, and Huang et al. [2025] construct hard perturbations on competition mathematics. Hao et al. [2025] stress-test mathematically equivalent transformations and observe sensitivity to variable naming and formatting. Our contribution is not a new perturbation. It is a controlled audit of what the existing ones measure once oracle reliability, difficulty matching, and item independence are handled.

Oracles and evaluation validity. The dependence of automated testing on a fallible oracle is well developed in software engineering [Barr et al., 2015]. Parallel concerns appear in NLP as

benchmark-validity and construct-validity critiques [Raji et al., 2021, Bowman and Dahl, 2021], which argue that a metric’s name frequently outruns what it measures. We instantiate that critique quantitatively for obfuscation benchmarks and show the induced bias is directional rather than random.

Contamination and proximity. Razeghi et al. [2022] show pretraining term frequency predicts arithmetic accuracy; Golchin and Surdeanu [2024] and Shi et al. [2024] develop membership-style detection. For closed-weight models no corpus ground truth exists, so we treat Infini-gram overlap [Liu et al., 2024] as an explicitly labelled public-corpus proximity proxy and use it only for within-family tests.

Chain-of-thought faithfulness. Turpin et al. [2023] and Lanham et al. [2023] intervene within a single reasoning trace. Our plan-execution measure instead compares a declaration made in one session against execution in a second session with no shared context, which isolates behavioural consistency from within-context self-consistency.

3 Setup

Problem families. We use three families chosen so that each controls a confound the others admit. **GSM** (44 items sampled from GSM8K [Cobbe et al., 2021] across Infini-gram proximity quartiles) provides an arithmetic family with documented contamination structure. **BW** (65 PlanBench instances [Valmeekam et al., 2023], 50 standard and 15 mystery) provides a planning family with a formal domain, which is what makes admissible-planner difficulty computation possible. **ALGO** (110 hand-designed dynamic-programming problems: coin change, shortest path, weighted interval scheduling) holds the underlying algorithm class fixed while varying public-corpus prevalence across roughly three orders of magnitude.

Variants. Each canonical item is expanded to six variants by deterministic scripts: W1 paraphrase, W2 reformat, W3 entity rename to a vocabulary-disjoint held-out set, W4 formal notation, W5 direction reversal, W6 procedural regeneration with the same algorithm and new parameters. Full protocol in Appendix A.

Validity gating. Two gates run before any model is queried. *Transform validity*: a variant is retained only if it differs from its canonical in the intended way. For W6 this means the gold answer must differ and the text must not be byte-identical; the gate excludes 33 rows with reason `variant_not_transformed`, of which 25 are ALGO interval rows whose rendered weights are unchanged under the parameter update the variant requests. *Oracle validity*: every gold answer is passed back through its own verifier (Section 4). Variants failing either gate are excluded with a recorded reason rather than scored as incorrect.

Models. Six models at $T=0$, zero-shot chain-of-thought via a single API gateway: Claude Sonnet 4, GPT-4o, Gemini 2.5 Flash, Llama-3.1-8B-Instruct [Dubey et al., 2024], o4-mini, and DeepSeek-R1-distill-Llama-70B [Guo et al., 2025] on the planning family.

Statistics. ALGO intervals are 10,000-draw percentile bootstraps resampling *clone families*, not individual items (Section 6); GSM and BW use Wilson intervals. Retention $R_{W3} = \text{Acc}_{W3} / \text{Acc}_{\text{can}}$ is reported only where $\text{Acc}_{\text{can}} \geq 0.30$, a threshold fixed before analysis, since the ratio is unstable near zero.

4 Verifier Design Determines the Measured Effect

Why obfuscation designs are the hard case. For a checker to score a renamed answer it must invert the rename. If the generator stores that mapping and the checker does not consume it, the checker cannot recognise a correct answer expressed in the renamed vocabulary. It then returns *incorrect* for every model on the perturbed condition while leaving the canonical condition untouched. The resulting scoring change is perfectly correlated with the experimental treatment, so it does not average out with more items or more models. It produces a clean, large, statistically significant drop that looks exactly like the hypothesis.

Table 1: Verifier ablations: measured accuracy with and without each verifier component, on identical model responses.

Component	Rows	Perturbed, on	Perturbed, off	Δ pert.	Δ canonical
Rename mapping (planning)	64	.252	.043	-.209	.000
Rename mapping (graph)	92	.537	.321	-.215	-.113
State parser	192	.397	.309	-.087	-.050

Table 2: Structural comparison of canonical versus regenerated Blocksworld instances ($n=47$ matched pairs; mystery instances excluded as they have no standard-domain encoding). Regenerated instances are harder or equal on every axis. Optimal plan length is computed with Fast Downward under an admissible heuristic.

Measure	Canonical	Regenerated	Δ
Blocks	6.19	6.45	+0.26
Goal clauses	5.19	6.23	+1.04
Goal tower depth	6.19	6.45	+0.26
Initial tower depth	1.02	1.00	-0.02
Optimal plan length	10.34	10.89	+0.55

Gold-in-gold-out. Before any model is queried, each item’s own gold answer is submitted to its verifier as though it were a model response. A verifier that rejects its own gold cannot score that cell. We run this for every (family, variant) pair and treat any failure as a blocking validity gate.

Verifier ablation. Holding model responses, items, and all other pipeline settings fixed, we toggle individual verifier components and remeasure reported accuracy (Table 1). Removing the planning rename-mapping consumer drops perturbed accuracy from .252 to .043 on 64/64 one-directional flips, with canonical delta .000 — the clean ablation of treatment-correlated sensitivity. The graph rename-mapping and state-parser toggles also move the perturbed condition, but they move the canonical condition as well, so each bundles treatment-correlated sensitivity with a change in general parsing coverage; we report them as such rather than as pure ablations of the treatment effect.

Implication. A published obfuscation result is only interpretable if the checker has been shown to accept its own gold under the perturbed condition. We recommend reporting gold-in-gold-out pass rates per (variant, family) cell alongside accuracy. In our pipeline this is a single table and adds negligible runtime.

5 Difficulty and Lexical Form Are Uncontrolled in Regenerated-Instance Comparisons

Comparing benchmark-released items against procedurally regenerated ones is the natural exposure control: same generator, same domain, no plausible pretraining exposure to the new items. The comparison is valid only under difficulty matching, which is rarely checked because most domains have no ground-truth difficulty measure. Planning does. Blocksworld has a formal PDDL encoding, so optimal plan length is computable with an admissible planner.

Naming confounds the unmatched gap. The unmatched Claude delta of +0.468 is not a clean difficulty effect. Canonical instances are overwhelmingly scattered-named (44/47) while regenerated instances are overwhelmingly sequential (37/47). Within naming strata the Claude deltas shrink to +0.090 (sequential) and +0.064 (scattered). On the $n=13$ pairs matched for naming convention, Claude moves only +0.077, GPT-4o +0.154, Gemini -0.154, and Llama/o4-mini/DeepSeek 0.000 (Table 4). We treat the naming imbalance as an identified confound, not as an estimated causal effect of sequential labels: the sequential-canonical cell is $n=3$.

Goal topology is outside current covariates. Goal topology requires a state-parser extension we have not yet validated; we report only the covariates we can currently compute (optimal plan length, block counts, naming convention, and the matched-naming stratum above).

Table 3: Accuracy on the same 47 matched pairs. Four of six models score higher on the structurally harder regenerated set.

Model	Canonical	Regenerated	Δ
Claude Sonnet 4	.170	.638	+.468
DeepSeek-R1-distill	.702	.936	+.234
GPT-4o	.085	.255	+.170
Llama-3.1-8B	.021	.021	.000
o4-mini	1.000	1.000	.000
Gemini 2.5 Flash	.489	.404	−.085

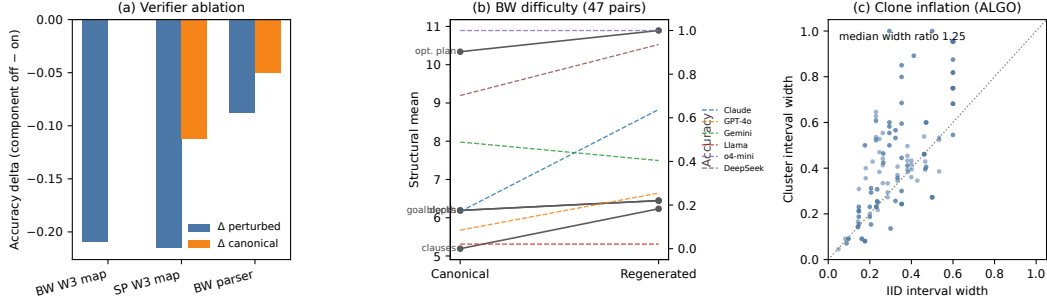


Figure 1: Three confounds that each produce drop-shaped signals. (a) Verifier ablation deltas on perturbed versus canonical conditions. (b) Blocksworld structural means (solid) rise under regeneration while accuracy (dashed) often rises too. (c) ALGO confidence-interval widths under iid versus clone-family cluster bootstrap.

Contamination does not account for the residual gap. If released instances were harder because models have memorised them in a way that interferes, per-instance proximity should predict the canonical-minus-regenerated gap. It does not: Spearman correlations between Infini-gram proximity and the per-instance gap have intervals spanning zero for five of six models (Claude -0.14 $[-0.40, 0.11]$; GPT-4o -0.11 ; Gemini -0.02 ; Llama -0.18 ; DeepSeek $+0.21$). Only o4-mini shows a significant negative association (-0.44 $[-0.60, -0.23]$, $p=0.001$). Restricting to GSM W3 retention among canonically correct items (N5_contamination_vs_retention.csv) likewise yields null associations for every model, with all confidence intervals spanning zero (Table 5).

The effect is domain-specific. Running the same comparison on the algorithmic family gives the opposite result: regenerated instances are harder in accuracy for four of five models (Claude $.667 \rightarrow .244$; o4-mini $1.000 \rightarrow .422$; Gemini $.500 \rightarrow .144$; GPT-4o $.433 \rightarrow .200$; Llama $.033 \rightarrow .078$), and are structurally mixed rather than uniformly harder. We therefore do not claim a general property of language models. We claim that a widely-used planning benchmark’s released instances are not interchangeable with instances its own generator produces once difficulty *and* lexical form are uncontrolled, and that any exposure comparison using released items inherits both confounds.

Direction reversal (W5) raises Blocksworld accuracy. Appendix A defines W5 as exchanging initial and goal states; the generator implements exactly that swap (swap_pddl_init_goal in variant_utils.py) and replans with Fast Downward. On the bank-restricted Probe 1 pool this is not a drop: Claude moves $.172 \rightarrow .873$, Gemini $.391 \rightarrow .764$, and o4-mini $.781 \rightarrow .909$ (Table 15). Goal/initial inversion is therefore a positive finding about surface-form sensitivity in planning, not an unexplained table anomaly. ALGO shortest-path W5 cells with majority parse_failed exclusions (GPT-4o 22/50, Gemini 32/50, o4-mini 8/50) are reported as exclusion reasons rather than accuracies.

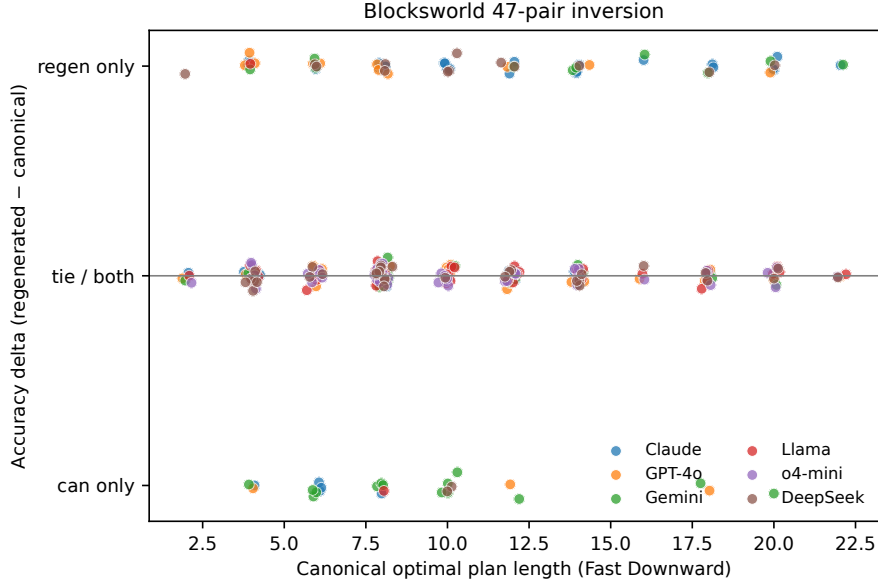


Figure 2: Centrepiece: 47 matched Blocksworld pairs. Each point is one (instance, model) with canonical optimal plan length on the x -axis and regenerated-minus-canonical correctness on the y -axis. Four of six models concentrate above zero despite harder structure.

Table 4: Blocksworld canonical vs regenerated accuracy stratified by block-naming convention (N1_bw_w6_stratified_accuracy.csv). Top: pairs matched on naming ($n=13$). Bottom: full naming $\times$ variant 2×2 with per-cell n . The sequential-canonical cell is $n=3$ and is reported only as a confound check, not as an effect estimate. Unmatched Claude $\Delta = +0.468$ shrinks to within-stratum $+0.090$ (sequential) and $+0.064$ (scattered).

Matched naming convention		Canonical	Regenerated	Δ
Model	n			
Claude	13	.231	.308	+.077
DeepSeek	13	.923	.923	.000
GPT-4o	13	.000	.154	+.154
Llama	13	.000	.000	.000
o4-mini	13	1.000	1.000	.000
Gemini	13	.385	.231	-.154
Naming $\times$ variant 2×2 (accuracy, n)				
Model	Seq./can.	Seq./W6	Scat./can.	Scat./W6
Claude	.667 ($n=3$)	.757 ($n=37$)	.136 ($n=44$)	.200 ($n=10$)
DeepSeek	.667 ($n=3$)	.946 ($n=37$)	.705 ($n=44$)	.900 ($n=10$)
GPT-4o	.000 ($n=3$)	.270 ($n=37$)	.091 ($n=44$)	.200 ($n=10$)
Llama	.000 ($n=3$)	.027 ($n=37$)	.023 ($n=44$)	.000 ($n=10$)
o4-mini	1.000 ($n=3$)	1.000 ($n=37$)	1.000 ($n=44$)	1.000 ($n=10$)
Gemini	.000 ($n=3$)	.459 ($n=37$)	.523 ($n=44$)	.200 ($n=10$)

6 Fragility Has No Item-Level Locus

Perturbation results are usually reported as marginal accuracies, which implicitly treats fragility as a property of items: some problems are harder to rename-survive than others. We test this directly by asking which items fail for *every* model.

No shared-hard set outside obfuscated planning. Among items where all five models were evaluated, the number failing for all of them is 0/110 on the algorithmic family and 0/20 on arithmetic. On planning it is 14/64, and all fourteen are mystery instances. Outside the obfuscated planning condition, essentially no item is universally fragile. Fragility is a model-item interaction, not an item property (Figure 3).

Table 5: GSM Infini-gram proximity vs W3 retention (N5_contamination_vs_retention.csv; canonical-correct only). All GSM intervals span zero.

Model	n	ρ	95% CI	p
Claude	37	-.089	[-.372, .198]	.561
GPT-4o	17	.158	[-.315, .583]	.537
Gemini	40	-.050	[-.350, .279]	.758
Llama	16	.191	[-.297, .611]	.461
o4-mini	37	.185	[-.063, .446]	.263

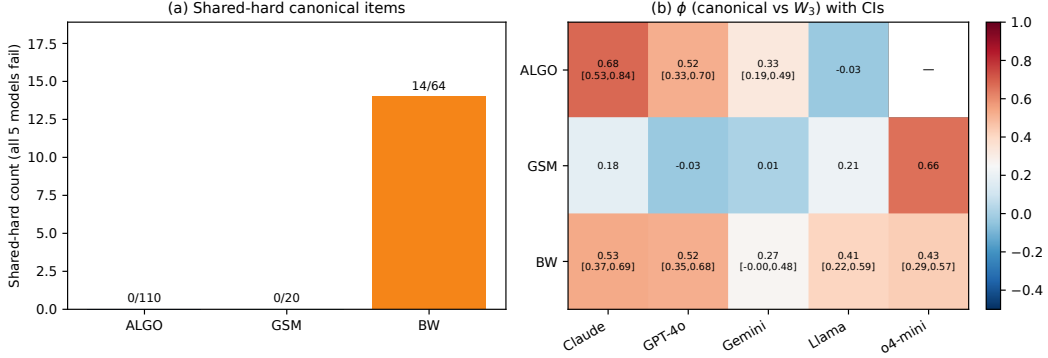


Figure 3: Item-level locus. (a) Shared-hard canonical counts (fail for all five paper models). (b) Canonical- W_3 contingency coefficient ϕ with bootstrap CIs.

Items are not independent. Near-duplicate detection with identical gold answers collapses the 110-problem algorithmic bank into 14 clone families covering 73 problems, for an effective n of 51; the largest family has 13 members. The frozen 61-item adversarial subset collapses to 14 clusters. Re-estimating intervals with a cluster bootstrap over families widens them by a median factor of 1.60 on the full bank and 1.94 on the adversarial subset, reaching 2.66 on individual cells. All ALGO intervals in this paper use the cluster bootstrap. Because procedural generators produce clone structure by construction, we expect this to generalise to other generated banks, and we recommend reporting effective n alongside nominal n .

Variant difficulty ordering is stable within family. Across models within a family, Kendall W on the W_1 - W_6 accuracy rank order is significant for ALGO ($W=0.419$, $p=0.049$), BW ($W=0.679$, $p<0.001$), and GSM ($W=0.660$, $p=0.042$) in P1_variant_ordering.csv (Table 6). The same ordering is *not* stable within model across families (Claude $W=0.165$, $p=0.824$), so “which surface transform is hardest” is a family property rather than a model trait.

Reliable crossovers exceed chance only on GSM. Pairwise model-item dissociation crossovers (C8_dissociation_crossovers.csv) tested against an additive item+model null (C8_null_expectation.csv) are at or below chance on ALGO and BW, but GSM reliable crossovers are above chance (20 observed vs null mean 0.57, $p=0.001$; Table 7). This is candidate behavioural double-dissociation evidence on arithmetic only; it does not generalise to the other families under the same null.

7 Convergent and Discriminant Validity of Robustness Measures

If surface invariance and procedural consistency were facets of one underlying robustness capacity, measures of each should covary (convergent validity) while remaining distinct from raw accuracy (discriminant validity). We test this with a measure that is methodologically independent of the re-name probe: a cross-session consistency index (CCI) comparing a model’s declared solution plan, produced in one session, against its step-by-step execution in a second session with no shared context.

Table 6: Stability of W1–W6 difficulty ordering (P1_variant_ordering.csv; Kendall W , 10k permutations).

Analysis	Scope	W	p	n rankers
within_family_across_models	ALGO	.419	.049	5
within_family_across_models	BW	.679	.000	6
within_family_across_models	GSM	.660	.042	3
within_model_across_families	Claude	.165	.824	3
within_model_across_families	Gemini	.238	.673	3
within_model_across_families	Llama	.793	.110	2
within_model_across_families	GPT-4o	.543	.462	2
within_model_across_families	o4-mini	.670	.043	3

Table 7: Reliable dissociation crossovers vs additive null (C8_null_expectation.csv; pairs in C8_dissociation_crossovers.csv).

Family	Observed	Null mean	$p(\geq \text{obs})$	Verdict
ALGO	0	8.79	1.000	at_or_below_chance
BW	3	4.14	.497	at_or_below_chance
GSM	20	.57	.001	above_chance

Pooled CCI covaries with rename survival. Pooling GSM with ALGO, the point-biserial correlation between CCI and rename survival is $r=0.243$ [0.027, 0.414], $p=0.024$, $n=272$; restricted to canonically correct items it is $r=0.256$ [0.043, 0.416], $p=0.020$, $n=178$ (Table 8). Both confidence intervals exclude zero. The p -values are two-sided cluster-bootstrap probabilities from the same 5k-resample procedure that produces the CIs (P2_P1_convergence.csv, seed 42).

Per-family estimates are underpowered relative to the pool. At the GSM item level alone the association is 0.156 [−0.006, 0.308], $p=0.060$, $n=128$ (Figure 4); ALGO alone is 0.359 [−0.043, 0.664], $p=0.085$, $n=144$. Across GSM models the rank correlation between mean CCI and retention is 0.600, $p=0.224$, $n=4$ (iid bootstrap). We read the per-family intervals that span zero as a power observation, not as evidence against the pooled association.

ALGO CCI. Mean plan–execution consistency on the algorithmic family (ALGO_P2_cci.csv) is Claude 0.384 ($n=59$ scored), GPT-4o 0.317 ($n=59$), and Llama 0.214 ($n=35$); Gemini contributes rows but no usable CCI scores (Table 9).

Variance decomposition. A crossed G-theory decomposition of binary accuracy (010_variance_components.csv, 010_generalizability_coefficients.csv) puts most ALGO primary-design variance on the model facet (proportion 0.327) rather than items (0.018), with $G_{\text{item}}=0.415$ (Table 10). GSM’s primary design instead shows a larger item share (0.186) and $G_{\text{item}}=0.680$. This is the variance-component backdrop for treating robustness measures as partially shared rather than identical to item difficulty.

Intrusion as a second behavioural measure. Among W3 errors, canonical-token intrusion rates (C1_intrusion_rates.csv) reach 0.619 for ALGO o4-mini and 0.250 for Claude, while BW W3 intrusion is 0 for every model with data (Table 11). Intrusion is therefore an independent behavioural signal that does not track rename accuracy alone.

Discriminant validity. Neither robustness measure tracks accuracy. Rank correlation between canonical accuracy and retention is 0.136 [−0.100, 0.800], $p=0.642$; between canonical accuracy and the canonical–rename contingency coefficient ϕ it is −0.426 [−0.724, 0.500], $p=0.146$. Robustness is measurably distinct from the quantity leaderboards report. We deliberately do not present retention and ϕ as convergent evidence: both are statistics of the same 2×2 contingency table, so their agreement is algebraic rather than empirical.

Compliance and correctness are separable. In a mid-solve injection protocol, models accept an injected incorrect intermediate state at high rates (Claude 88.5%, GPT-4o 93.4%, o4-mini 100%,

Table 8: CCI vs rename survival point-biserial (P2_P1_convergence.csv). CIs and p from the same cluster bootstrap (seed 42, 5k resamples).

Scope	r	95% CI	p	n
GSM_all_instances_with_p2	.156	[-.006,.308]	.060	128
GSM_canonical_correct_subset	.153	[-.030,.320]	.109	110
ALGO_all_instances_with_p2	.359	[-.043,.664]	.085	144
ALGO_canonical_correct_subset	.300	[-.061,.676]	.119	68
GSM_ALGO_pooled_all_instances_with_p2	.243	[-.027,.414]	.024	272
GSM_ALGO_pooled_canonical_correct_subset	.256	[-.043,.416]	.020	178

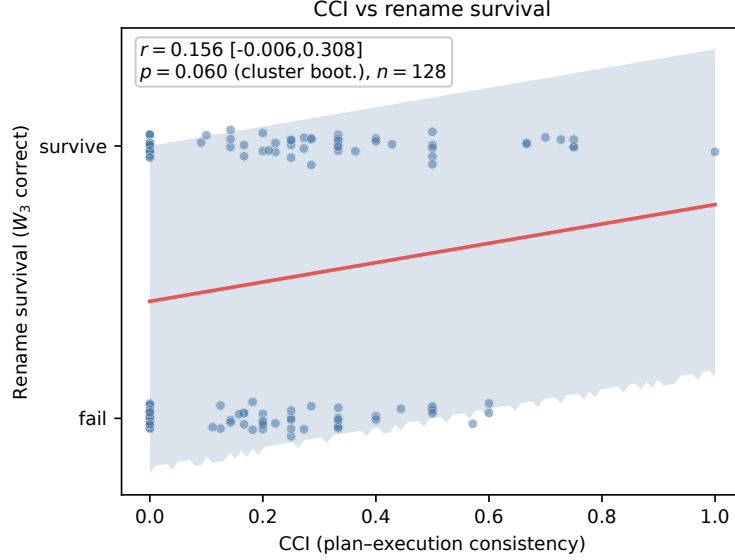


Figure 4: CCI versus rename survival on GSM ($n=128$), with OLS fit and residual bootstrap band. Annotated $r=0.156$ and $p=0.060$ are the cluster-bootstrap point-biserial (P2_P1_convergence.csv, 5k resamples, seed 42). The pooled GSM+ALGO association is $r=0.243$, $p=0.024$, $n=272$ (Table 8).

Llama 39.3%; Gemini treats every injection as malformed input and proceeds without it) yet post-injection final accuracy stays near uninterrupted rates (Claude .525 vs .541; GPT-4o .508 vs .557; o4-mini .377 vs .409). Accepting a false premise mid-derivation does not reliably derail the final answer, so compliance is not a proxy for procedural integrity.

8 Limits of the Present Design

8.1 Model-imposed limits

Under a strict PDDL execution protocol, planning sessions abort on illegal actions at high rates; routing through a natural-language tolerant parser shifts the failure mode from format errors to precondition violations (BW_P2_floor_documentation.csv: goal-reached rate 0 for Claude/GPT-4o/Llama on $n=50$ sessions each; mean CCI 0.013/0.000/0.109). Models emit genuinely illegal plans rather than mis-formatted valid ones, and BW Probe 2 CCI is an execution floor rather than a model fingerprint. The Phase 2A (uninjected CCI) vs Phase 2B (injected TEP) protocol split is documented in P2_phase2A_2B_protocol.csv.

The ALGO logit-lens rank-shift vs W3-correct analysis (N3_algo_mech_behavior_link.csv) is **NOT ESTIMABLE**: Llama has 1/60 W3-correct outcomes and Qwen2.5-1.5B has 0/61 (Table 12). Outcome degeneracy means the present open-weight sample cannot support a within-model correlation between representational shift and rename survival. Separately, planning gold answers use three

Table 9: ALGO Probe 2 CCI (ALGO_P2_cci.csv). Gemini has no usable CCI scores in this file.

Model	Mean CCI	n scored	n rows
Claude	.384	59	61
GPT-4o	.317	59	61
Gemini	—	0	61
Llama	.214	35	61

Table 10: Primary G-theory variance components (O10_variance_components.csv) and coefficients (O10_generalizability_coefficients.csv).

Family	Design	Prop. model	Prop. item	G_{item}	ϕ_{item}
ALGO	max_cells	.327	.018	.415	.018
BW	full_7_variants	.383	.002	.125	.002
GSM	full_7_variants	.009	.186	.680	.186

distinct first-action tokens across 65 problems, so rank-based readouts over the gold token cannot discriminate conditions in that family.

8.2 Measurements pending inference budget

ALGO shortest-path W5 still has majority parse_failed cells for GPT-4o (22/50), Gemini (32/50), and o4-mini (8/50). GSM GPT-4o and Llama Probe 1 cells are limited to $n=20$ included rows per variant under api_error exclusions (24 of 64 attempted, alongside missing_bank_row). Phase-resolved execution accuracy on arithmetic is not separable in the current logs because the implementation pools the two sessions. A difficulty-and-naming-matched regenerated Blocksworld bank is specified and estimated at under 100 calls once the generator match criteria are locked.

8.3 Design scope

Closed-model corpus membership is unresolvable for Claude, GPT-4o, Gemini, o4-mini, and DeepSeek (O16_CLOSED_MODEL_CAVEAT.md); Infini-gram proxy scores remain available but are not calibratable against known membership. Infini-gram windows are not cross-family comparable ($n \leq 8$ for arithmetic, $n \leq 13$ otherwise; P3_infinigram_query_audit.csv). Clone families reduce effective n on ALGO (Section 6) and all ALGO intervals use the cluster bootstrap for that reason. Released-versus-regenerated comparisons require matching on covariates that are not obvious *a priori*; we identify two—structural difficulty and block-naming convention—and leave goal topology outside the covariates we can currently compute (Section 5).

We fit single-class and mixture Rasch models on GSM (C2_irt_fit_comparison.csv); BIC selects $K=1$ (mixture Rasch BIC 558.1), and strategy_posteriors.csv assigns all 72 GSM person-rows to a single mixed class at MAP probability 1.0 (Table 13). ALGO and BW were not fit: O10 item variance fell below the pre-registered gate. Per-instance triple-probe labels remain underdetermined across a 270-configuration threshold sweep (retrieval-consistent counts from 0 to 170 of 440), so we report probe-level results rather than conjunction labels.

9 Conclusion

Perturbation benchmarks measure a real phenomenon, but the protocol as commonly implemented conflates it with at least three others. Verifier design determines the measured drop under obfuscation. Regenerated and released instances differ in difficulty *and* lexical form; on Blocksworld the unmatched accuracy inversion shrinks once naming is stratified, and goal topology sits outside the covariates we can currently compute. Generated banks contain clone families that halve the effective sample size. Once these are handled, fragility turns out to have no item-level locus outside obfuscated planning, and plan-execution consistency shows pooled convergent association with rename survival while remaining discriminant from raw accuracy — so reporting robustness as one number still outruns what any single cell measures.

Table 11: Canonical-token intrusion among W3 errors (C1_intrusion_rates.csv).

Family	Model	n errors	Intrusion rate	95% CI
ALGO	Claude	12	.250	[.00,.55]
ALGO	GPT-4o	37	.054	[.00,.19]
ALGO	Gemini	28	.143	[.03,.39]
ALGO	Llama	48	.000	[.00,.00]
ALGO	o4-mini	21	.619	[.33,.87]
BW	Claude	58	.000	[.00,.00]
BW	GPT-4o	54	.000	[.00,.00]
BW	Gemini	56	.000	[.00,.00]
BW	Llama	57	.000	[.00,.00]
BW	o4-mini	32	.000	[.00,.00]

Table 12: Mechanistic rank-shift vs W3 correctness (N3_algo_mech_behavior_link.csv). Outcome degeneracy makes ρ not estimable.

Model	n	W3-correct	Status
Llama-3.1-8B	60	1/60	NOT ESTIMABLE
Qwen2.5-1.5B	61	0/61	NOT ESTIMABLE

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Table 13: Mixture IRT / Rasch model comparison (C2_irt_fit_comparison.csv). Strategy posteriors in strategy_posteriors.csv ($n=72$ GSM rows, $K=1$).

Family	Model	K	BIC	Selected	Note
GSM	1PL	1	476.1	no	baseline single-class Rasch/1PL
GSM	2PL	1	449.3	no	baseline 2PL
GSM	mixture_rasch	2	587.8	no	restarts=4; converged_iter 65
GSM	mixture_rasch	3	617.5	no	restarts=4; converged_iter 9
GSM	mixture_rasch	4	641.8	no	restarts=4; converged_iter 41
GSM	mixture_rasch	1	558.1	yes	SELECTED by BIC ($K \in 1..4$); LRT-path $K=1$; single-class mixture
ALGO	NOT_FIT	—	—	no	Skipped: O10 item_var=0.0051 prop=0.0182 below gate (var>=0.
BW	NOT_FIT	—	—	no	Skipped: O10 item_var=0.0006 prop=0.0024 below gate (var>=0.

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A Variant Generation Protocol

W1 (paraphrase). Restated in different words; numeric values, entities, and structural relations preserved exactly. **W2 (format).** Prose converted to a structured representation appropriate to the family (adjacency matrix, interval table, equation set). **W3 (entity rename).** All entity identifiers replaced with a held-out, vocabulary-disjoint token set, length-matched and single-token where the original was single-token. The generator persists the mapping; the verifier consumes it (Section 4). **W4 (formal notation).** L^AT_EX for arithmetic, formal predicates for planning, mathematical edge-list notation for graph and interval problems. **W5 (direction reversal).** Planning: initial and goal states exchanged. Arithmetic: forward problems inverted to backward form. Graph: source and destination exchanged. Not defined for coin change or interval scheduling, which have no meaningful reversal. **W6 (procedural regeneration).** Same algorithm and parameter ranges, new numeric values. A regenerated item is retained only if its gold answer differs from the canonical and its text is not byte-identical.

Every variant passes its family verifier on the gold answer before any model is queried.

B Coverage and Exclusions

Full per-(family, slice, model, variant) Probe 1 accuracies with CIs are in Table 15 (tables/table_variants.tex), regenerated from rescored included=True rows consistent with P1_rescore_summary.csv.

C Metric Definitions

$R_{W3} = \text{Acc}_{W3} / \text{Acc}_{\text{can}}$, reported only where $\text{Acc}_{\text{can}} \geq 0.30$. ϕ is the contingency coefficient of canonical and rename correctness over items. CCI is the fraction of declared numeric steps matched by second-session execution within $\epsilon=0.01$. Error propagation is the fraction of post-injection steps differing from the uninjected run. Proximity is Infini-gram n -gram overlap against public corpora, used only within family; window sizes differ across families (maximum $n=8$ for arithmetic, 13 for the others) and the count is unnormalised, so cross-family comparison is not valid.

D Protocol and Contamination Audit Notes

Infini-gram query construction. P3_infinigram_query_audit.csv records that GSM uses full problem text with $\max n=8$, while ALGO/BW default to $\max n=13$ on index v4_rpj_llama_s4, so contamination scores are not comparable across families. ALGO “template”/“instance” columns share the full problem text under different n caps (template forced through the GSM $n=8$ path; instance uses the gold string), so they are not a true template/instance split. BW instance fallback blocksworld num_blocks N is a keyword dump when no goal snippet is present.

BW Probe 2 floor. BW_P2_floor_documentation.csv documents goal-reached rate 0 for all three models with data and places BW CCI/TEP at an execution floor; those series are not used as model fingerprints in the main text.

Phase 2A/2B separation. P2_phase2A_2B_protocol.csv lists the scripts, output files, and whether injection lands at critical_step_index for ALGO, GSM, and BW. ALGO 2A/2B are separate process invocations; GSM stores CCI and TEP on one row; BW CCI and TEP use distinct scripts.

Closed-model contamination caveat. As stated in 016_CLOSED_MODEL_CAVEAT.md, exact/near-exact corpus membership is identifiable only for open-corpus models (Pythia/Pile, OLMo/Dolma). For Claude, GPT-4o, Gemini, o4-mini, and DeepSeek the ground-truth membership cell cannot be constructed; Infini-gram proxy scores remain available but are not calibratable against known membership.

Table 14: Coverage and exclusions from COVERAGE_MASTER.csv. One row per (probe, family, model, variant).

Probe	Family	Model	Variant	Att.	Incl.	Excl.	Top exclusion
P1	ALGO	Claude	W1	110	110	0	—
P1	ALGO	Claude	W2	110	110	0	—
P1	ALGO	Claude	W3	110	100	10	parse_failed
P1	ALGO	Claude	W4	110	108	2	parse_failed
P1	ALGO	Claude	W5	50	50	0	—
P1	ALGO	Claude	W6	90	90	0	—
P1	ALGO	Claude	canonical	110	110	0	—
P1	ALGO	Gemini	W1	110	110	0	—
P1	ALGO	Gemini	W2	110	110	0	—
P1	ALGO	Gemini	W3	110	105	5	parse_failed
P1	ALGO	Gemini	W4	110	105	5	parse_failed
P1	ALGO	Gemini	W5	50	32	18	parse_failed
P1	ALGO	Gemini	W6	90	86	4	parse_failed
P1	ALGO	Gemini	canonical	110	110	0	—
P1	ALGO	Llama	W1	110	110	0	—
P1	ALGO	Llama	W2	110	104	6	parse_failed
P1	ALGO	Llama	W3	110	109	1	parse_failed
P1	ALGO	Llama	W4	110	106	4	parse_failed
P1	ALGO	Llama	W5	50	50	0	—
P1	ALGO	Llama	W6	110	90	20	missing_bank_row
P1	ALGO	Llama	canonical	110	110	0	—
P1	ALGO	mock	W1	1	0	1	mock_row
P1	ALGO	mock	W1	1	0	1	mock_row
P1	ALGO	mock	W2	1	0	1	mock_row
P1	ALGO	mock	canonical	1	0	1	mock_row
P1	ALGO	mock	canonical	1	0	1	mock_row
P1	ALGO	GPT-4o	W1	110	110	0	—
P1	ALGO	GPT-4o	W2	110	110	0	—
P1	ALGO	GPT-4o	W3	110	102	8	parse_failed
P1	ALGO	GPT-4o	W4	110	110	0	—
P1	ALGO	GPT-4o	W5	50	22	28	parse_failed
P1	ALGO	GPT-4o	W6	110	90	20	missing_bank_row
P1	ALGO	GPT-4o	canonical	110	110	0	—
P1	ALGO	o4-mini	W1	110	109	1	parse_failed
P1	ALGO	o4-mini	W2	110	105	5	parse_failed
P1	ALGO	o4-mini	W3	110	81	29	parse_failed
P1	ALGO	o4-mini	W4	110	97	13	parse_failed
P1	ALGO	o4-mini	W5	50	8	42	parse_failed
P1	ALGO	o4-mini	W6	90	89	1	parse_failed
P1	ALGO	o4-mini	canonical	110	110	0	—
P1	BW	The answer is 42.	MOCK	4	0	4	mock_row
P1	BW	Claude	W1	109	102	7	invalid_gold
P1	BW	Claude	W2	109	103	6	invalid_gold
P1	BW	Claude	W3	109	104	5	invalid_gold
P1	BW	Claude	W4	109	103	6	invalid_gold
P1	BW	Claude	W5	109	99	10	invalid_gold
P1	BW	Claude	W6	89	84	5	invalid_gold
P1	BW	Claude	canonical	109	108	1	invalid_gold
P1	BW	DeepSeek	W1	65	58	7	invalid_gold
P1	BW	DeepSeek	W2	65	59	6	invalid_gold
P1	BW	DeepSeek	W3	65	60	5	invalid_gold
P1	BW	DeepSeek	W4	65	59	6	invalid_gold
P1	BW	DeepSeek	W5	65	55	10	invalid_gold
P1	BW	DeepSeek	W6	65	60	5	invalid_gold
P1	BW	DeepSeek	canonical	49	48	1	invalid_gold
P1	BW	Gemini	W1	65	58	7	invalid_gold
P1	BW	Gemini	W2	65	59	6	invalid_gold
P1	BW	Gemini	W3	65	60	5	invalid_gold
P1	BW	Gemini	W4	65	59	6	invalid_gold
P1	BW	Gemini	W5	65	55	10	invalid_gold
P1	BW	Gemini	W6	65	60	5	invalid_gold
P1	BW	Gemini	canonical	65	64	1	invalid_gold
P1	BW	Llama	W1	109	102	7	invalid_gold
P1	BW	Llama	W2	109	103	6	invalid_gold
P1	BW	Llama	W3	109	103	6	invalid_gold
P1	BW	Llama	W4	109	102	7	invalid_gold
P1	BW	Llama	W5	109	99	10	invalid_gold
P1	BW	Llama	W6	104	84	20	missing_bank_row
P1	BW	Llama	canonical	109	108	1	invalid_gold
P1	BW	mock	W1	1	0	1	mock_row
P1	BW	mock	W2	11	0	11	mock_row
P1	BW	mock	W3	10	0	10	mock_row
P1	BW	mock	W4	10	0	10	mock_row
P1	BW	mock	W5	9	0	9	mock_row
P1	BW	mock	canonical	17	0	17	mock_row
P1	BW	GPT-4o	W1	109	102	7	invalid_gold
P1	BW	GPT-4o	W2	109	103	6	invalid_gold
P1	BW	GPT-4o	W3	109	104	5	invalid_gold
P1	BW	GPT-4o	W4	109	103	6	invalid_gold
P1	BW	GPT-4o	W5	109	99	10	invalid_gold
P1	BW	GPT-4o	W6	104	84	20	missing_bank_row
P1	BW	GPT-4o	canonical	109	108	1	invalid_gold
P1	BW	o4-mini	W1	65	58	7	invalid_gold
P1	BW	o4-mini	W2	65	59	6	invalid_gold
P1	BW	o4-mini	W3	65	60	5	invalid_gold

Table 15: Per-variant accuracy by family, slice, and model (Probe 1). Accuracies use rescored included=True rows consistent with results/derived/P1_rescore_summary.csv. ALGO intervals are 10k cluster-bootstrap CIs over clone families (seed 42); GSM/BW use Wilson 95% CIs. Excluded cells show the exclusion reason, never 0.

Family	Slice	Model	Can.	W1	W2	W3	W4	W5	W6
GSM	–	Claude	.841 [.71,.92]	.841 [.71,.92]	.773 [.63,.87]	.750 [.61,.85]	.636 [.49,.76]	.818 [.68,.90]	.750 [.55,.88]
GSM	–	GPT-4o	.850 [.64,.95]	.750 [.53,.89]	.300 [.15,.52]	.300 [.15,.52]	.200 [.08,.42]	.300 [.15,.52]	<i>missing_bank_row</i>
GSM	–	Gemini	.909 [.79,.96]	.818 [.68,.90]	.636 [.49,.76]	.523 [.38,.66]	.477 [.34,.62]	.614 [.47,.74]	.958 [.80,.99]
GSM	–	Llama	.800 [.58,.92]	.850 [.64,.95]	.250 [.11,.47]	.150 [.05,.36]	.300 [.15,.52]	.050 [.01,.24]	<i>missing_bank_row</i>
GSM	–	o4-mini	.841 [.71,.92]	.864 [.73,.94]	.818 [.68,.90]	.841 [.71,.92]	.682 [.53,.80]	.886 [.76,.95]	.833 [.64,.93]
ALGO	CC-chall	Claude	.700 [.00,.95]	.700 [.00,.88]	.700 [.00,.95]	.600 [.00,.75]	.800 [.00,1.00]	<i>not defined</i>	<i>variant_not_transformed</i>
ALGO	CC-chall	GPT-4o	.600 [.00,.75]	.400 [.00,.55]	.600 [.00,.82]	.600 [.00,.82]	.500 [.00,.68]	<i>not defined</i>	<i>missing_bank_row</i>
ALGO	CC-chall	Gemini	.500 [.00,.68]	.700 [.00,.95]	.600 [.00,.82]	.700 [.00,.95]	.700 [.00,.95]	<i>not defined</i>	<i>variant_not_transformed</i>
ALGO	CC-chall	Llama	.200 [.00,.27]	.100 [.00,.14]	.400 [.00,.75]	.000 [.00,.00]	.222 [.00,.32]	<i>not defined</i>	<i>missing_bank_row</i>
ALGO	CC-std	Claude	.267 [.07,.47]	.467 [.20,.73]	.067 [.00,.23]	.300 [.00,.56]	.667 [.44,.87]	<i>not defined</i>	.067 [.00,.23]
ALGO	CC-std	GPT-4o	.267 [.08,.44]	.400 [.21,.60]	.000 [.00,.00]	.071 [.00,.20]	.867 [.64,1.00]	<i>not defined</i>	.200 [.00,.43]
ALGO	CC-std	Gemini	.267 [.07,.47]	.133 [.00,.36]	.000 [.00,.00]	.000 [.00,.00]	.267 [.07,.47]	<i>not defined</i>	.267 [.07,.50]
ALGO	CC-std	Llama	.000 [.00,.00]	.067 [.00,.23]	.111 [.00,.30]	.000 [.00,.00]	.000 [.00,.00]	<i>not defined</i>	.067 [.00,.23]
ALGO	SP-chall	Claude	.941 [.69,1.00]	.941 [.81,1.00]	.676 [.35,.92]	.882 [.63,.98]	.941 [.69,1.00]	.000 [.00,.00]	.323 [.25,.36]
ALGO	SP-chall	GPT-4o	.412 [.04,.89]	.529 [.29,.89]	.147 [.00,.26]	.294 [.04,.69]	.676 [.24,.95]	.000 [.00,.00]	.258 [.25,.27]
ALGO	SP-chall	Gemini	.706 [.41,.93]	.471 [.11,.77]	.235 [.04,.50]	.353 [.15,.78]	.912 [.77,.97]	.100 [.00,.25]	.129 [.00,.18]
ALGO	SP-chall	Llama	.029 [.00,.10]	.147 [.00,.31]	.029 [.00,.07]	.000 [.00,.00]	.088 [.00,.15]	.000 [.00,.00]	.129 [.00,.25]
ALGO	SP-std	Claude	.952 [.83,1.00]	1.000 [1.00,1.00]	.762 [.54,.95]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	.053 [.00,.16]	.579 [.33,.80]
ALGO	SP-std	GPT-4o	.762 [.54,.95]	.667 [.44,.86]	.048 [.00,.14]	.476 [.29,.67]	.714 [.50,.89]	.000 [.00,.00]	.368 [.11,.63]
ALGO	SP-std	Gemini	.619 [.41,.82]	.762 [.57,.92]	.810 [.60,1.00]	.476 [.22,.71]	1.000 [1.00,1.00]	.083 [.00,.23]	.333 [.00,.65]
ALGO	SP-std	Llama	.048 [.00,.14]	.143 [.00,.27]	.000 [.00,.00]	.000 [.00,.00]	.143 [.00,.29]	.053 [.00,.16]	.105 [.00,.24]
ALGO	WIS-chall	Claude	.353 [.00,.60]	.176 [.00,.30]	.118 [.00,.50]	.000 [.00,.00]	.067 [.00,.60]	<i>not defined</i>	<i>variant_not_transformed</i>
ALGO	WIS-chall	GPT-4o	.353 [.00,.41]	.176 [.00,.24]	.000 [.00,.00]	.000 [.00,.00]	.000 [.00,.00]	<i>not defined</i>	<i>no included</i>
ALGO	WIS-chall	Gemini	.353 [.00,.60]	.176 [.00,.24]	.000 [.00,.00]	.000 [.00,.00]	.000 [.00,.00]	<i>not defined</i>	<i>variant_not_transformed</i>
ALGO	WIS-chall	Llama	.059 [.00,.08]	.000 [.00,.00]	.000 [.00,.00]	.059 [.00,.08]	.000 [.00,.00]	<i>not defined</i>	<i>no included</i>
ALGO	WIS-std	Claude	.077 [.00,.23]	.231 [.00,.46]	.231 [.00,.46]	.091 [.00,.27]	.077 [.00,.23]	<i>not defined</i>	<i>variant_not_transformed</i>
ALGO	WIS-std	GPT-4o	.154 [.00,.38]	.231 [.00,.46]	.000 [.00,.00]	.000 [.00,.00]	.000 [.00,.00]	<i>not defined</i>	<i>no included</i>
ALGO	WIS-std	Gemini	.000 [.00,.00]	.000 [.00,.00]	.231 [.00,.46]	.182 [.00,.45]	.000 [.00,.00]	<i>not defined</i>	<i>variant_not_transformed</i>
ALGO	WIS-std	Llama	.000 [.00,.00]	.000 [.00,.00]	.000 [.00,.00]	.077 [.00,.23]	.000 [.00,.00]	<i>not defined</i>	<i>no included</i>
BW	–	Claude	.172 [.10,.28]	.138 [.07,.25]	.305 [.20,.43]	.100 [.05,.20]	.119 [.06,.23]	.873 [.76,.94]	.673 [.54,.78]
BW	–	GPT-4o	.063 [.02,.15]	.172 [.10,.29]	.102 [.05,.20]	.100 [.05,.20]	.119 [.06,.23]	.291 [.19,.42]	.250 [.15,.38]
BW	–	Gemini	.391 [.28,.51]	.259 [.16,.38]	.237 [.15,.36]	.133 [.07,.24]	.068 [.03,.16]	.764 [.64,.86]	.404 [.28,.54]
BW	–	Llama	.016 [.00,.08]	.052 [.02,.14]	.017 [.00,.09]	.050 [.02,.14]	.000 [.00,.06]	.000 [.00,.07]	.019 [.00,.10]
BW	–	o4-mini	.781 [.67,.86]	.862 [.75,.93]	.831 [.72,.91]	.483 [.36,.61]	.475 [.35,.60]	.909 [.80,.96]	1.000 [.93,1.00]